

Probability distribution function

Discrete Random Variable

- Takes separate, distinct values
- usually integers
- Comes from counting

Example

No. of students in class

No. of emails received

Continuous Random Variable

- Take infinitely many values
- Can be decimal / Fractional
- Comes from measurement

Examples

Height of a person

Weight

0

IF it is measured $\rightarrow$ Continuous

IF it is counted $\rightarrow$ Discrete

Probability Distribution Function



Probability Density
Function

eg Continuous Random
variable

Probability Mass
Function

eg Discrete Random
variable

Random Variable - It is just a numeric outcome of some random process.

OR

A Random variable X is a real number which is determined by the outcomes of

the random experiment

eg

Tossing 2 coins simultaneously
sample space = $\{HH, HT, TH, TT\}$

Let the random variable be getting
number of heads then

$$X(s) = \{0, 1, 2\}$$

* Types of Random Variables

- 1) Discrete Random Variable
- 2) Continuous Random Variable

* Discrete Probability Distribution

Let x is a Discrete Random Variable
with possible outcomes $x_1, x_2, x_3, \dots, x_n$
having probabilities $P(x_i)$ for $i = 1, 2, \dots, n$
IF $P(x_i) > 0$ and $\sum_{i=1}^n P(x) = 1$ then the

Function $P(x_i)$ is called **Probability Mass Function** of a random variable X and $\{x_i, P(x_i)\}$ for $i=1, 2, \dots, n$ is called **Discrete Probability Distribution**

eg

Tossing 2 Coins Simultaneously
sample space = $\{HH, HT, TH, TT\}$

Let the random variable be getting number of heads then

$$X(S) = \{0, 1, 2\}$$

$$\text{Probability of getting no heads} = \frac{1}{4}$$

$$\text{Prob. of getting 1 head} = \frac{2}{4} = \frac{1}{2}$$

$$\text{Prob of getting 2 heads} = \frac{1}{4}$$

$\therefore$ Discrete probability distribution

x_i	0	1	2
$P(x_i)$	$1/4$	$1/2$	$1/4$

→ Random variable

→ Prob mass function

Cumulative Distribution Function is given by

$$F(x) = P[X \leq x] = \sum_{i=0}^x P(x_i)$$

eg

- 1) IF 3 cars are selected randomly from 6 cars having 2 defective cars
- a) Find prob distribution of defective cars.
- b) Find the expected number of defective cars.

sol

Number of ways to select 3 cars from 6 cars $= {}^6C_3$

$$\frac{6 \times 5 \times 4}{2 \times 1} = 20$$

$$\overline{3 \times 2 \times 1}$$

Let Random Variable = No. of defective cars
 $X(S)$

$$\text{Prob of non defective cars} = \frac{{}^4C_2 {}^2C_0}{{}^6C_3} = \frac{1}{5}$$

$$\text{Prob of one defective cars} = \frac{{}^4C_2 {}^2C_1}{{}^6C_3} = \frac{3}{5}$$

$$\text{Prob of 3 defective cars} = \frac{{}^4C_1 {}^2C_2}{{}^6C_3} = \frac{1}{5}$$

clearly $P(x_i) > 0$ & $\sum_{i=1}^n P(x_i) = 1$

$\therefore$ Prob distribution of defective car is

x_i	0	1	2
$P(x_i)$	$1/5$	$3/5$	$1/5$

Permutation

It is an arrangement of items where order matters.

Formula

$${}^n P_r = \frac{n!}{(n-r)!}$$

$${}^6 P_3 = \frac{6 \times 5 \times 4 \times \cancel{3 \times 2 \times 1}}{\cancel{3 \times 2 \times 1}} = 120$$

Examples

Ranking (1st, 2nd, 3rd)

Passwords/PINs

Order of task

Seating arrangement

Combination

It is a selection of items where order

does not matter

Formula

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

$${}^6C_r = \frac{6 \times 5 \times 4 \times \cancel{3 \times 2 \times 1}}{\cancel{3 \times 2 \times 1} \times 3 \times 2 \times 1} = 20$$

Examples

Selecting Teams

Lottery numbers

selecting cards from deck

Probability Problems

$$\text{mean}(\mu) = \sum x_i \times P(x_i) = E(X)$$

$$\text{Variance} = \sum (x_i)^2 \times P(x_i) - \mu^2$$

$$\boxed{\sum_{i=1}^n P(x_i) = 1}$$

* Continuous Probability Distribution

Let X be continuous random variable taking values on the interval (a, b) . A function $f(x)$ is said to be **Probability density function** of x if

1) $f(x) > 0 \quad \forall x \in (a, b)$

2) Total area under the probability curve is 1 i.e. $\int_a^b f(x) dx = 1$

3) For two distinct numbers 'c' and 'd' in (a, b) is given by $P(c < x < d) =$
Area under the probability curve
between ordinates $x = c$ & $x = d$ i.e.
 $\int_c^d f(x) dx.$